

ECON485 - Homework Assignment 3

JOIN Queries

Task 1: List Students and Registered Sections

This query displays a list of students together with the course code and section number of their registered sections.

```
SELECT s.FullName, c.CourseCode, sec.SectionNumber
FROM Students s
JOIN Registrations r ON s.StudentID = r.StudentID
JOIN Sections sec ON r.SectionID = sec.SectionID
JOIN Courses c ON sec.CourseID = c.CourseID;
```

Task 2: Courses and Total Students

This query calculates the total number of students registered for each course. Using LEFT JOIN ensures that courses with 0 students are also displayed.

```
SELECT c.CourseCode, COUNT(r.StudentID) AS TotalStudents
FROM Courses c
LEFT JOIN Sections sec ON c.CourseID = sec.CourseID
LEFT JOIN Registrations r ON sec.SectionID = r.SectionID
GROUP BY c.CourseCode;
```

Task 5: Detect Prerequisite Violations (Advanced JOIN)

This query identifies students who are registered for a course without satisfying its prerequisites. It checks for students who either have no record of completing the prerequisite (NULL grade via LEFT JOIN) or whose grade falls below the required threshold.

```
SELECT
    st.FullName AS StudentName,
    c.CourseCode AS TargetCourse,
    req_c.CourseCode AS MissingPrerequisite,
    cc.Grade AS StudentGrade,
    p.MinimumGrade
FROM Registrations r
JOIN Sections sec ON r.SectionID = sec.SectionID
JOIN Courses c ON sec.CourseID = c.CourseID
JOIN Students st ON r.StudentID = st.StudentID
JOIN Prerequisites p ON c.CourseID = p.CourseID
JOIN Courses req_c ON p.RequiredCourseID = req_c.CourseID
LEFT JOIN CompletedCourses cc ON p.RequiredCourseID = cc.CourseID
                                AND st.StudentID = cc.StudentID
WHERE cc.Grade IS NULL OR cc.Grade < p.MinimumGrade;
```